Evolution and Unified Performance Evaluation of Clustering Techniques in Heterogeneous Wireless Sensor Networks

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***Abstract*—Wireless Sensor Networks (WSNs) face strict energy limits, so effective communication protocols are crucial to extend network life and ensure steady data transmission. Among various strategies, clustering based routing stands out for cutting communication costs and evenly distributing energy use across nodes. In the last 20 years, these methods have advanced from simple probabilistic and threshold driven schemes to optimisation and machine learning approaches [2], [3]. A comprehensive survey of hierarchical clustering evolution confirms that LEACH based enhancements continue to serve as the dominant baseline for IoT oriented WSN protocol design Clustering is the standard response [1] [2]. We conduct a chronological performance analysis of LEACH, PEGASIS, TEEN, APTEEN, NSGA II, Fuzzy T2, Kohonen SOM, DQN (RL based), and GCN under a standardised Python setup. This implements the first order radio model, uniform parameters, and metrics like FND, HND, PDR, residual energy, throughput, energy efficiency ratio, number of alive nodes, and communication overhead for rigorous, fair benchmarking. Outcomes trace an upward trend: as clustering grows adaptive and learning based, so do longevity, balance, and reliability. Learning protocols like DQN and GCN shine in lifetime extension and energy preservation, while static probabilistic ones fail sooner due to fixed cluster head logic. Overall, the analysis maps clustering's development and equips designers for next generation efficient WSNs.**

***Index Terms*—Wireless Sensor Networks, Clustering Protocols, Network Lifetime, Energy Efficiency, Reinforcement Learning, Graph Neural Networks, Fuzzy Logic T2, Cluster Head Selection**

I. Introduction

A wireless sensor node spends most of its energy on its radio, and in nearly all deployments its battery cannot be replaced [1]. Lifetime is therefore decided by how the network organizes its transmissions rather than by any property of the sensing hardware [1]. Clustering is the standard response: nodes elect a subset of themselves as cluster heads, members send short-range packets to their head, the head fuses them into one packet and pays the single expensive transmission to the base station. Because that role is expensive, it has to rotate, and the question every clustering protocol answers is which nodes should carry it this round.

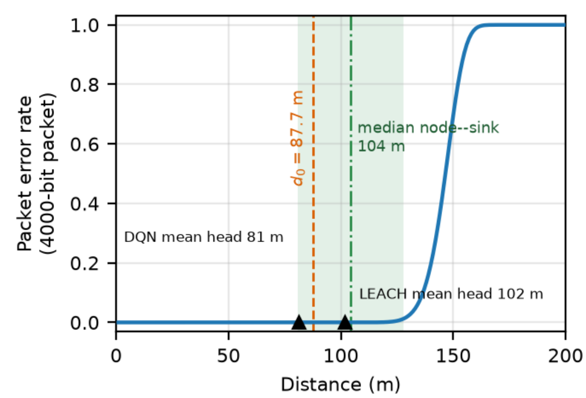

Fig. 1. 1

Twenty-five years of answers fall into three broad generations. The first is heuristic. LEACH rotates the role by an independent probabilistic threshold with an epoch constraint [8]; PEGASIS abandons clusters entirely for a greedy chain in which each node talks only to its nearest neighbour [10]; TEEN and its extension APTEEN leave the topology alone and instead suppress transmissions that fall below a sensed-value threshold [11], [12]. The second generation replaces the heuristic with an explicit objective. Cluster-head selection is posed as a multi-objective optimization problem and solved with NSGA-II [16] , or encoded in a type-2 fuzzy rule base that reasons over residual energy, distance and local density under linguistic uncertainty [19], [20]. The third generation learns the selection function instead of specifying it [3]: a self-organizing map places heads by unsupervised vector quantization [25], a deep Q-network treats rotation as a sequential decision problem [27], [28], and a graph convolutional network scores candidates from the topology's adjacency structure [29], [30].

The generations are usually compared by reading their published tables side by side, and that comparison does not hold. Each protocol was proposed against a different simulator, field size, node count, packet length and energy budget [4], [5], almost always against LEACH alone, and almost always on a lossless channel. More consequentially, the assumptions that decide the outcome are rarely stated. Whether a centralized protocol is charged for the uplink that tells the base station each node's residual energy can be worth a quarter of the entire energy budget over a long run. Whether fusion is lossless in bits determines how much a chain protocol gains over a clustered one. Whether lifetime is reported at first death, half death or last death can reverse a ranking outright, and the author chooses which to report. A margin obtained under one set of these choices is not evidence about the algorithm; it is partly evidence about the harness.

This paper measures the three generations in one harness. Nine protocols spanning all of them, plus a direct-transmission baseline, run on a single engine over 30 paired trials in each of two channel configurations, for 600 runs in total. Run index i seeds the topology, the per-link shadowing and the entire sensed-value stream identically for every protocol, so a comparison at a given index is genuinely paired rather than an average over unrelated worlds. Fairness is enforced structurally instead of by convention: the engine owns energy accounting through a single charging function, packet sizing, the aggregation charge, the channel and its ARQ retries, and the counting of delivered readings. A protocol returns only a cluster structure and a route shape, and a static audit verifies that no protocol implementation writes to the energy or liveness arrays. Every protocol runs on the same numerical substrate, hand-written NumPy including the backpropagation for both learned models, because per-round setup time is a reported metric and mixing frameworks would corrupt it. No protocol is tuned to reproduce any published number; disagreement with the source tables is expected and is not treated as error.

Four results are worth stating up front. First, transmit power is not a background parameter but a determinant of the ranking. Repeating the entire study on a lossless channel removes the first-death advantage of the type-2 fuzzy protocol and the DQN over LEACH altogether — $+417$ and $+426$ rounds at $p < 0.001$ under packet loss, falling to $+17$ and $+22$ rounds and no significance once loss is removed. Direct measurement of where the energy goes explains it: LEACH's average cluster head sits at 101.8 m, past the $d_0 \approx 87.7$ m crossover onto the fourth-power branch and far enough out to lose packets, and spends 6.4% of its budget on retransmissions, against 2.3% for the DQN at 81.1 m. What the newer protocols demonstrably do well is keep the head-to-sink hop inside the free-space regime, which pays only in proportion to how lossy that hop is. Second, the effect is specific to first death; differences in area under the alive-node curve and in readings delivered hold at $p < 0.001$ in both channels. Third, no single lifetime figure summarizes clustering, because it buys a later first death by bringing the last death forward: LEACH reaches first death at 1046 rounds against the baseline's 114, a factor of 9.2, while dying out completely at 2191 rounds against the baseline's 3202. The GCN is the extreme case, with a first death 3.1× earlier than LEACH's and a last death 45% later. Fourth, the reactive protocols survive by not reporting: TEEN and APTEEN fail to reach last death within the 7000-round cap in all 30 trials, at a data yield of 0.11–0.12 against roughly 0.9 for the proactive protocols.

The contributions are as follows.

1. A single simulation engine in which nine clustering protocols from three methodological generations execute under provably identical physical conditions, with fairness enforced by construction rather than by author discipline. 2. A paired experimental design — common topology, shadowing and sensed-data stream per trial index — analysed with paired sign-flip permutation tests, paired bootstrap intervals and Holm–Bonferroni correction across the protocol family, in place of the unpaired mean-and-deviation comparison the literature normally reports. 3. A channel sensitivity pair, run as a rule rather than an appendix: a conclusion that survives both the lossy and the ideal configuration is a statement about the protocol, and one that flips is a statement about transmit power. Applying the rule retracts a comparison that the lossy channel alone would have supported. 4. Mechanistic explanations obtained by measurement rather than argument, including the energy split by category, cluster-head distance relative to the radio crossover, and cluster-head rotation concentration. 5. A full disclosure of the modelling choices that bound these results, ordered by how far each could move a conclusion, together with the reproduction artifacts.

Section II describes the system and radio model, Section III the nine protocols and their fairness-relevant implementation details, Section IV the experimental design and statistical procedure, Section V the results, Section VI the threats to validity, and Section VII concludes.

— Two things I flag for you rather than decide silently.

Contribution 3 is the paper's real novelty and I'd argue it should also appear in the abstract — "we retract one of our own comparisons" is unusual and reviewers reward it. But it sets up an expectation that Section VI delivers the full A2 analysis, not a summary.

The intro currently claims the centralized-subsidy problem in the framing ("whether a centralized protocol is charged for the uplink… can be worth a quarter of the entire energy budget") without saying that we adopt the piggyback assumption too. That's honest as written — it's motivation, not a claim of immunity — but Section VI must own it, since the measured 12.73% vs 2.33–3.95% control-energy gap means the cross-class numbers carry a 9–10 point subsidy. If you want the intro to disclose it directly instead, say so and I'll add a sentence; it costs a little rhetorical momentum but closes the gap between paragraph three and our own method.

II. System Model

A. Network and Energy

One hundred nodes are placed uniformly at random in a 100×100 m field. The base station sits outside the field at $(50, 150)$, so node-to-sink distances span 50 to 158 m. Every node begins with $E_0 = 1$ J and is stationary for

TABLE I

SIMULATION PARAMETERS. EVERY VALUE IS READ FROM THE SINGLE FROZEN CONFIGURATION OBJECT USED BY ALL TEN PROTOCOLS; NOTHING IS PROTOCOL-SPECIFIC.

Parameter	Value
Topology and energy	
Nodes N	100
Field	100 × 100 m
Base station	$(50, 150)$ m
Initial energy E_0	1.0 J per node
Rounds (cap)	7,000
Runs per configuration	30
Packets	
Data packet	4,000 bits
Control packet	200 bits
Radio (first-order model)	
E_{elec}	50 nJ/bit
ε_{fs}	10 pJ/bit/m^2
ε_{mp}	0.0013 pJ/bit/m^4
E_{DA} (fusion)	5 nJ/bit
Crossover d_0	87.7 m
Channel	
Transmit power P_t	12 dBm
Path loss at 1 m	40 dB
Path-loss exponent n	3.0
Shadowing σ	4 dB
Noise floor	-105 dBm
ARQ retransmissions	2
Sensed data (Ornstein–Uhlenbeck)	
Mean μ, reversion θ	50, 0.05
Step σ	2.0
TEEN/APTEEN H_T, S_T, T_C	55, 1.0, 50 rounds
Protocol-agnostic	
Head fraction p	0.05
Density radius	25 m
NSGA-II population, generations	40, 30
Re-clustering interval	5 rounds
Timing	
Bit rate	250 kbps
Slot / guard / processing	20 / 4 / 1 ms

the life of the network; nodes fail only by energy depletion. The configuration carries fields for two-level and three-level heterogeneity, but they are disabled throughout this study, and every result reported here is for a homogeneous population.

A run proceeds in rounds. In each round every living node takes a sensor reading, the protocol under test decides the cluster structure, and the resulting transmissions are executed by the engine. A run terminates when the last node dies or at a cap of 7000 rounds, whichever comes first.

Energy is a linear reservoir with a single charging function. When a node is asked to spend more than it holds, it spends what remains, dies, and the packet in flight is lost. The alternative — refusing the transmission and stranding the remainder — was rejected because it breaks the conservation identity

$$\sum_i E_0^{(i)} - \sum_i E_{\text{res}}^{(i)} = E_{\text{tx}} + E_{\text{rx}} + E_{\text{ctrl}} + E_{\text{agg}} + E_{\text{retry}}$$

which is asserted at the end of every run to 10^{-9} J and is the primary correctness check on the whole simulator.

B. Radio Energy

We use the first-order model of Heinzelman et al. [1]. Transmitting k bits over distance d costs

$$E_{\text{tx}}(k,d) = \begin{cases} kE_{\text{elec}} + k\varepsilon_{fs}d^2, & d < d_0 \;\; kE_{\text{elec}} + k\varepsilon_{mp}d^4, \\ d \geq d_0 \end{cases}$$

with $E_{\text{rx}}(k) = kE_{\text{elec}}$ and a fusion charge of E_{DA} per bit received. The crossover $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}} = 87.7$ m sits inside the range of node-to-sink distances, which is what makes cluster-head placement relative to d_0 a live variable rather than a formality — a head at 100 m pays the fourth-power branch, one at 80 m does not.

The model is deliberately minimal and its omissions matter. There is no idle-listening cost, no sleep-wake transition energy, no radio startup, and no processor energy. In real deployments idle listening often dominates the budget, and including it would compress the differences between all nine protocols. We note this here rather than only in the limitations, because it bounds the interpretation of every energy figure that follows.

C. Channel

Packet loss is derived from a link budget rather than assigned as a constant. Path loss follows a log-distance law with per-link shadowing,

$$PL(d) = PL_{1m} + 10n\log_{10} d + X_\sigma, \qquad X_\sigma \sim \mathcal{N}(0, \sigma^2)$$

giving a receive power of $P_t - PL(d)$ and an SNR against a -105 dBm noise floor. Bit errors follow the closed-form non-coherent BFSK expression $\text{BER} = \frac{1}{2}e^{-\gamma/2}$, and packet error follows $\text{PER} = 1 - (1 - \text{BER})^k$, evaluated in the numerically stable form $-\text{expm1}(k\ln(1-\text{BER}))$. Non-coherent BFSK was chosen over BPSK because it is closed-form and needs no error function, allowing the entire channel to be vectorized without a SciPy dependency; since setup time is a reported metric, avoiding a heavyweight dependency in the inner loop is a fairness consideration as much as a convenience.

The resulting curve is a waterfall, not a gradient. Without shadowing, PER for a 4000-bit packet is 2.6×10^{-8} at 100 m, 1.0×10^{-3} at 120 m, 0.19 at 140 m and 0.97 at 158 m: the transition from usable to useless spans about 3.6 dB of SNR and roughly 40 m of distance. Log-normal shadowing at $\sigma = 4$ dB is what smears that step into a distribution over links, and it is the source of the genuine per-topology variance in delivery ratio. Shadowing is drawn once per topology and held fixed, because the nodes are static; re-drawing it per packet would model fast fading, which averages out over 7000 rounds and would contribute nothing but noise.

Failed packets are retried up to twice, for three attempts in total, and each attempt pays full transmit and receive energy. This is the mechanism that makes a lossy channel cost joules rather than only packets, and it is charged to a separate accounting category so that retry energy can be measured directly rather than inferred.

Because path loss, shadowing and packet size are all fixed once the topology is built, PER and transmit energy are constant per link. Both are precomputed as lookup tables at construction and indexed in the round loop. This is a pure cache — the values are identical to recomputing them — and it consumes no random draws, so it cannot perturb the seeding order described in Section IV.

D. Control-Traffic Accounting

Control traffic is charged by the engine according to the protocol's declared class, not by the protocol itself, and both endpoints pay.

Distributed protocols pay a full handshake every round: the head broadcasts an advertisement priced at the distance to its farthest joiner and every living non-head pays a receive charge per advertisement heard, each member sends a join request and the head receives it, and the head then broadcasts a TDMA schedule that every member receives. The chain protocol pays no handshake at all, but passes a leader token hop by hop along the chain, with sender and receiver charged on each link, plus a single sink broadcast on rounds where the chain is rebuilt.

Centralized protocols skip the advertisement and join-request steps, since the sink has already told each node its assignment, and pay a single sink broadcast on re-clustering rounds. They do still pay the per-round TDMA schedule broadcast under identical pricing to the distributed case, including the same skip for heads with no members. Their uplink node state — residual energy and position-derived features — is treated as piggybacked on data traffic the network already sends and is charged nothing.

That last choice is the single largest modelling decision in the paper and we state its magnitude rather than only its existence. A separate 200-bit status report per node per round, over a sink distance above 100 m and therefore on the fourth-power branch, costs about 3.6×10^{-5} J per node per round, which accumulates to roughly a quarter of the entire 1 J budget across the round cap. Charging it would sink all five centralized protocols for a reason unconnected to the quality of their clustering. Not charging it grants them an advantage we measure in Section VI and must be read alongside every cross-class comparison.

E. Sensed Data

Each node's reading follows a discrete Ornstein–Uhlenbeck process,

$$v(r) = v(r{-}1) + \theta\,(\mu - v(r{-}1)) + \mathcal{N}(0, \sigma^2)$$

with a stationary distribution of approximately $\mathcal{N}(50, 6.4^2)$. A pure random walk was rejected: over 7000 rounds it drifts by roughly 167 units and pegs against any clipping bound, which would make the reactive protocols' behaviour an artifact of the clip rather than of their thresholds. The threshold parameters of TEEN and APTEEN are set against this stationary distribution — a hard threshold at $+0.78\sigma$ and a soft threshold at 0.16σ — and were not adjusted after observing any result. The full sensed stream for all rounds is generated once per topology, so every protocol reads byte-identical sensor values.

F. Latency

Latency is analytic rather than queueing-based:

$$L = s \cdot T_{\text{slot}} + h \cdot (T_{\text{tx}} + T_{\text{proc}})$$

where s is the sender's TDMA slot index and h its hop depth. Slot length is derived from the bit rate rather than assumed, at $4000/250{,}000 = 16.0$ ms of transmit time plus a 4 ms guard. A cluster head's own packet carries a slot index equal to its member count, since it can forward only after every member slot has elapsed; this is what makes cluster latency scale with cluster size. The chain protocol uses a slot index of zero and accumulates delay through hop depth instead. There is no queueing delay, no MAC contention, and retransmissions cost energy but not modelled time, so the absolute figures are lower bounds and only the ordering should be read.

G. Engine-Enforced Invariants

Three rules are enforced by the engine and cannot be bypassed by a protocol implementation. First, transmissions within a round execute sorted by hop depth and then by node index, not by node index alone; plain index ordering breaks causality whenever a forwarding node's index falls below one of its senders', which occurs for roughly half of all head-member pairings on a random topology and silently corrupts fusion totals. A runtime check raises if any node forwards while inbound packets remain pending. Second, a node that receives data may emit at most one outgoing data packet and pays the fusion charge on every bit it received. Third, the count of original readings represented in a delivered packet is computed by the engine from a boolean flag on each transmission, not declared by the protocol; an earlier version allowed protocols to declare it, which would have let a protocol inflate its own delivery figures.

— Table I in the file lists all parameters with values: $n = 100$, field 100×100 m, BS $(50, 150)$, $E_0 = 1$ J, $p = 0.05$, data 4000 b, control 200 b, $E_{\text{elec}} = 50$ nJ/b, $\varepsilon_{fs} = 10$ pJ/b/m², $\varepsilon_{mp} = 0.0013$ pJ/b/m, $E_{\text{DA}} = 5$ nJ/b, $d_0 = 87.7$ m, $P_t = 12$ dBm, $PL_{1m} = 40$ dB, $n = 3.0$, $\sigma = 4$ dB, noise floor -105 dBm, 2 retries, $\theta = 0.05$, $\mu = 50$, $\sigma_{\text{sense}} = 2.0$, HT $= 55$, ST $= 1.0$, TC $= 50$, 250 kb/s, 4 ms guard, 1 ms processing, cap 7000 rounds, 30 runs.

One correction to what I told you last turn. In the introduction commentary I wrote that centralized protocols skip the uplink charge, which is right, but I should have been precise that they still pay the TDMA schedule broadcast every round under the same pricing as LEACH. The subsidy is narrower than "centralized protocols pay almost no control traffic" — it is specifically the advertisement, the join request, and the uplink report. That is why their measured control energy is 2.3–4.0

Fig. 1 should be the PER-vs-distance curve with the d_0 line and the node-to-sink distance band marked. It carries §II-C on its own and justifies the sensitivity pair before the reader reaches Section IV.

III. Protocols Under Comparison

Nine protocols are implemented, spanning three methodological generations, plus a direct-transmission baseline in which every node sends its own reading straight to the sink. The baseline exists to establish what clustering is worth at all, and it turns out to be a more informative control than expected (Section V).

All nine share one structural contract. Each targets the same nominal cluster count $k = \max(1, \text{round}(p \cdot N_{\text{alive}}))$, recomputed each round from the living population so that k decays as the network dies. Membership is always nearest-head by Euclidean distance. What each protocol returns to the engine is a set of head identities and a membership map, nothing else; energy, packet sizing, fusion charges, the channel, and reading accounting are the engine's, as described in Section II-G. Realized head counts are not forced to match the target, because clamping LEACH's stochastic election would be implementing a different algorithm.

A. Generation One: Heuristics

LEACH [1] implements the paper's election threshold unmodified. A node that has not yet been a head in the current epoch self-elects with probability $T(n) = p/(1 - p(r \bmod \lfloor 1/p \rfloor))$, and the "already served" set clears every $\lfloor 1/p \rfloor = 20$ rounds, dropping any node that has died meanwhile. This is worth stating precisely because the epoch mechanism turns out to be LEACH's real strength: it guarantees each surviving node serves exactly once per epoch, which is close to optimal for the first-death metric specifically, and no other protocol here has an equivalent guarantee.

PEGASIS [2] abandons clusters. A greedy chain is built from the node farthest from the sink, repeatedly appending the nearest unchained living node. Since positions are static, the only event that can invalidate a chain is a death, so the chain is rebuilt exactly when the living count drops. The leader rotates round-robin over chain positions. Data flows inward from both ends toward the leader and fuses at every intermediate hop, so each non-leader emits exactly one packet and the entire network's readings reach the sink in a single transmission. PEGASIS is the sole multi-hop protocol in the study and the sole beneficiary of the ideal-fusion assumption at full scale.

TEEN [3] subclasses LEACH and reuses its clustering unchanged. This is not an implementation shortcut but the correct reading of the protocol: TEEN's contribution is entirely the reporting rule, and holding clustering fixed makes the comparison a controlled one. A node transmits only when its reading exceeds a hard threshold and has moved by at least a soft threshold since its last actual transmission. APTEEN [4] adds the periodic half — a node also transmits if it has been silent for a count-time interval — so a node that never crosses the hard threshold still reports at least once every 50 rounds. Only the reporting-frequency mechanism is modelled; APTEEN's attribute-based and historical query facilities are out of scope and are declared as such.

B. Generation Two: Explicit Optimization

NSGA-II [5] poses head selection as a genuine multi-objective problem, solved with the full algorithm — fast non-dominated sort, crowding distance, binary tournament selection, and elitist $(\mu + \lambda)$ survival — not a weighted sum standing in for it. A genome is a vector of k distinct living node indices, so cardinality is fixed by construction and no repair operator is required. Three objectives are minimized: the total energy of one round under the candidate assignment, the negated minimum residual energy among selected heads, and the standard deviation of cluster membership counts.

The third objective was chosen against the more common intra-cluster distance variance, for a reason worth recording: distance already dominates the first objective through the d^2 and d^4 terms, and a second strongly correlated distance objective would collapse the Pareto front toward a line and waste the multi-objective machinery. Membership-count spread is genuinely orthogonal — it measures load balance rather than geometry. The first objective is computed by calling the engine's own radio functions, so the search optimizes exactly the cost model it will later be charged against rather than an approximation of it. Selection from the final front is by knee point: each objective is normalized across the front and the individual nearest the ideal point is taken. This is deterministic and introduces no hand-tuned weights, which would have been a back door to calibrating against published results.

Interval type-2 fuzzy selection scores every living node from residual energy, distance to sink, and local density, with Gaussian membership functions of uncertain mean, a 10% footprint of uncertainty, Karnik–Mendel iterative type reduction, and midpoint defuzzification [6]. The 27-rule base is generated by score $= 2E + (2 - D) + D_n$ over three-level inputs, giving a table monotone in every input by construction. Monotonicity is what makes the rule base defensible without tuning; the double weight on residual energy is a designer's choice and we label it as one rather than presenting it as derived.

C. Generation Three: Learned Selection

SOM [7] trains a Kohonen map with k units on a one-dimensional chain neighbourhood over normalized position and residual energy, one hundred iterations with geometrically decaying learning rate and radius, retrained from scratch on the same five-round cadence as NSGA-II and the fuzzy system. The node nearest each converged unit becomes a head. It is the least sophisticated of the three learned methods and, as Section V shows, the one whose weakness is most legible.

DQN [8] and GCN [9] are deliberately built to isolate a single difference. Both consume the identical four normalized features — residual energy, distance to sink, local density, and rounds since last serving as head — with the GCN importing the DQN's feature function rather than redefining it. Both use hand-written forward and backward passes verified against central finite differences, and both are trained on seeds 100–129, frozen, and evaluated on the disjoint seeds 0–29. What differs is only the architecture and the training signal.

The DQN is a $4 \rightarrow 16 \rightarrow 2$ multilayer perceptron trained with experience replay, a periodically synchronized target network, and $\gamma = 0.999$. At evaluation it does not let each node act on its own greedy action, which would

TABLE II

THE TEN CONFIGURATIONS COMPARED. "CONTROL MODEL" DETERMINES WHICH SETUP MESSAGES ARE CHARGED: DISTRIBUTED PROTOCOLS PAY ADVERTISEMENT, JOIN AND SCHEDULE; THE CHAIN PROTOCOL PAYS A HOP-BY-HOP TOKEN; CENTRALIZED PROTOCOLS PAY ONLY THE DOWNLINK SCHEDULE, THEIR UPLINK STATE BEING PIGGYBACKED ON DATA TRAFFIC (SECTION **??**).

Protocol	Gen.	Head-selection rule	Control model	Re-cluster	Offline train
Direct (no clustering)	–	None	None	–	No
LEACH	G1	Randomised rotation	Distributed	Every round	No
PEGASIS	G1	Greedy chain	Chain	On death	No
TEEN	G1	LEACH + hard/soft threshold	Distributed	Every round	No
APTEEN	G1	TEEN + count-down timer	Distributed	Every round	No
NSGA-II	G2	Multi-objective search (3 obj.)	Centralized	Every 5 rounds	No
Fuzzy T2	G2	Interval type-2 rule base (27 rules)	Centralized	Every round	No
SOM	G3	Self-organising map	Centralized	Every round	Yes
DQN	G3	Learned Q-value ranking	Centralized	Every round	Yes (frozen)
GCN	G3	Learned graph scoring	Centralized	Every round	Yes (frozen)

make the head count fluctuate anywhere between zero and the entire population; instead nodes are ranked by the advantage $Q_{\mathrm{CH}} - Q_{\neg\mathrm{CH}}$ and the top k are selected. This makes it a learned ranking function rather than a set of independent agents, which is a real narrowing of the reinforcement-learning framing and is documented as such rather than glossed over.

The GCN is two graph-convolutional layers of width 16 over a symmetrized 6-nearest-neighbour graph with the standard symmetric-normalized propagation rule, trained self-supervised against a differentiable proxy for energy balance. It is deliberately not trained to imitate a heuristic — doing so would make any subsequent comparison against that heuristic vacuous — and shares no code path with the DQN's training loop. Selection is greedy top-k by score subject to a 15 m minimum separation, and if separation exhausts the candidate pool, fewer than k heads are accepted rather than padding the set back up.

The contrast this construction buys is the paper's cleanest: with identical inputs, the DQN has temporal credit assignment and the GCN does not. The GCN can see rotation state through its fourth feature but nothing in its single-round loss rewards acting on it. Section V measures what that costs.

D. Disclosure on the DQN's Hyperparameters

One protocol's parameters were revised after observing a failure, and we state it rather than let it be inferred. The initial configuration used $\gamma = 0.95$, an effective horizon of about twenty rounds against a first-death timescale three orders of magnitude longer, and it diverged. The fix raised γ to 0.999, normalized the return magnitude, and extended training from 30 to 100 episodes. The reward function itself — round energy, residual-energy spread across living nodes, and a death penalty — was not changed.

At that point a stopping rule was fixed in advance: the result would be recorded as final regardless of outcome, with no further changes to reward, architecture, features, or hyperparameters. It was then evaluated on five held-out seeds, beating random head rotation on all five. Stating the pre-commitment is what makes that number credible; without it, "we fixed the DQN until it worked" and "we fixed a divergence and stopped" are indistinguishable from the outside.

— Table II in the file lists the ten configurations against class, control-traffic model, re-clustering cadence, and whether pre-training is used.

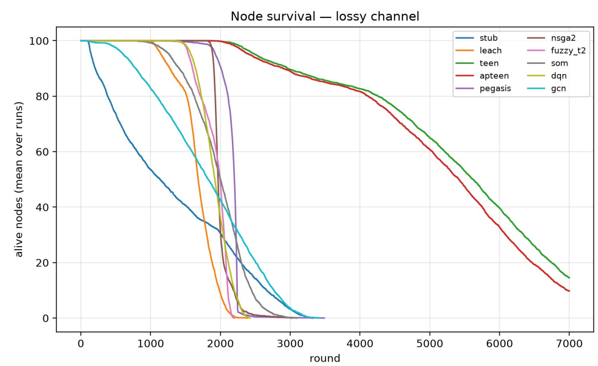

Fig. 2. 1

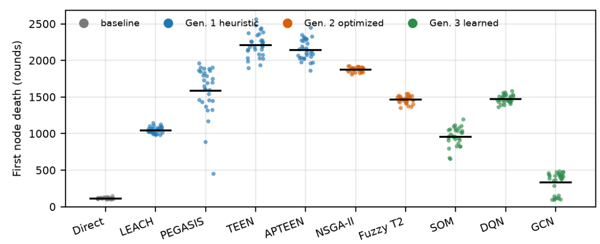

Fig. 3. 1

IV. Experimental Design

A. Paired Trials

Every protocol is evaluated over 30 trials in each channel configuration, for 600 runs in total. The trials are paired rather than independent. Trial index i seeds a single random stream that generates, in fixed order, the node positions, the initial energies, the full per-link shadowing matrix, and the sensed-value array for all rounds. That order depends only on i and never on which protocol is being run, so at a given index all ten configurations face a byte-identical physical world. A protocol cannot see a different topology by virtue of being a different protocol.

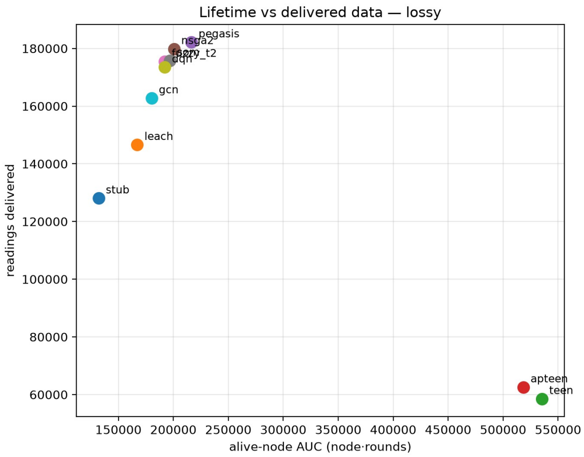

Fig. 4. 1

TABLE III
DELIVERY, LOSSY CHANNEL. PACKETS AT THE BASE STATION MEASURE AGGREGATION DEPTH, NOT DELIVERY, AND ARE NOT COMPARABLE ACROSS PROTOCOLS; READINGS DELIVERED AND DATA YIELD ARE. DATA YIELD IS READINGS ARRIVING AT THE SINK DIVIDED BY READINGS TAKEN, SO IT IS THE COLUMN THAT PRICES TEEN AND APTEEN'S SUPPRESSION.

Protocol	Gen.	Packets at BS ($\times 10^3$)	Readings delivered ($\times 10^3$)	PDR (%)	Data
Direct (no clustering)	–	128.1	128.1	96.9	[illegible]
LEACH	G1	11.0	146.6	87.7	[illegible]
PEGASIS	G1	2.3	182.2	84.2	[illegible]
TEEN	G1	21.2	58.6	90.8	[illegible]
APTEEN	G1	21.5	62.6	90.8	[illegible]
NSGA-II	G2	9.7	179.9	89.6	[illegible]
Fuzzy T2	G2	9.1	175.4	91.4	[illegible]
SOM	G3	9.4	175.8	89.3	[illegible]
DQN	G3	9.0	173.5	90.3	[illegible]
GCN	G3	8.6	162.8	90.3	[illegible]

Protocol-internal randomness — LEACH's election draws, the genetic operators, the ϵ-greedy exploration during training — comes from a separate stream derived by a keyed hash of the run index and protocol name. Python's built-in hash is salted per process and is never used anywhere in the simulator, since it would silently destroy reproducibility across invocations.

The pairing is what licenses the statistical procedure in §IV-D, and it is a substantially stronger design than the unpaired mean-and-standard-deviation comparison this literature normally reports. Topology variance is large relative to the differences between protocols, and an unpaired test spends most of its power fighting variance that a paired design eliminates by construction.

One thing is deliberately not shared: the per-packet channel draws. Different protocols make different numbers of draws per round, so there is no coherent sense in which two protocols could experience the same coin flips packet by packet. The fairness-critical quantity is the static per-link shadowing, which is identical. These are distinct claims — that the physical channel geometry is fair, and that the exact sequence of random outcomes is fair — and only the first needs to hold.

B. The Channel Sensitivity Pair

The entire study is run twice, once with the packet-error model of §II-C enabled and once with it disabled. This is not a robustness appendix but a decision rule fixed before the results were read:

A conclusion that holds in both channel configurations is a statement about the protocol. A conclusion that changes between them is a statement about transmit power.

Transmit power is a parameter the reference specifications do not fix, and §II-C showed that the packet-error curve is a waterfall whose position determines how severely single-hop protocols are penalized on the sink hop. Choosing that parameter to make a metric discriminating would be precisely the calibration this study forbids. Running both endpoints and reporting only what survives removes the choice from the argument. Section V reports one conclusion that the rule retracts.

C. Metrics

Lifetime is reported at three points — first, half, and last node death — plus the area under the living-node curve, which is the integral of surviving nodes over rounds. All four are reported together because the three death points do not agree on an ordering, for reasons Section V makes concrete. Runs that reach the round cap without full depletion are flagged as censored, and a censored last-death figure is a lower bound rather than a measurement; it is never averaged into a mean without the censoring count reported beside it.

Delivery is reported in two units, and the distinction matters more than it appears to. Packets received at the sink measures aggregation depth, not delivery: one chain round delivers a single fused packet where one clustered round delivers roughly five, so on a packet basis the chain protocol appears five times worse while actually delivering more sensor readings. Original readings represented in delivered packets is the cross-protocol comparable quantity, and the packet count is reported only as an aggregation-ratio indicator.

The same defect once affected the delivery ratio, which is why it is now reading-based. The packet-based formula reported ten percent on a lossless channel, purely because ten source readings became one aggregated packet. Suppressed readings are not counted as losses in either numerator or denominator, since a reading that was never injected was never lost; the cost of suppression appears instead in data yield, defined as readings delivered over readings taken, which is the metric that legitimately penalizes the reactive protocols.

Computational cost is reported in three separate forms that are never merged: measured wall-clock setup time and peak memory, counted algorithmic operations, and analytical complexity. Wall-clock reflects implementation quality as much as algorithm design; counted operations are more honest, and analytical complexity is the most honest of the three.

D. A Measurement Artifact Worth Reporting

Timing and memory are measured in separate runs, and the reason generalizes beyond this study. Allocation tracing intercepts every allocation and its overhead grows with the number of traced blocks, so measuring both in one run inflated setup time by factors between 3.5 and 7.8 depending on how many small objects a protocol allocated. The self-organizing map, which allocates roughly forty thousand temporary arrays per retraining, was hit hardest; the vectorized heuristics barely at all.

Because the inflation was non-uniform, it distorted the ranking rather than only the scale. The GCN's true setup cost is about 35% higher than LEACH's, but under tracing it measured about 30% lower — the sign of the comparison reversed. The resolution is that trial zero of each protocol measures memory and reports no timing, while the remaining trials do the reverse. Peak memory therefore comes from a single sample and has no defined deviation, and timing means are over $n-1$ trials. Simulation outcomes are unaffected, since tracing changes wall-clock only and never energy or delivery. We report this because any comparative study that profiles memory and time simultaneously is silently ranking its subjects by allocation count.

E. Statistical Procedure

Because trials are paired, each comparison against a reference protocol reduces to a vector of 30 per-topology differences. The null hypothesis that the difference distribution is symmetric about zero is tested by a sign-flip permutation test: 20,000 random sign vectors are applied to the observed differences, and the p-value is the proportion of permuted means at least as extreme as the observed one, with the standard add-one correction to both numerator and denominator so that no p-value is ever reported as exactly zero. The floor at 30 paired samples is therefore approximately 5×10^{-5}, and comparisons at that floor are reported as such rather than as zero.

The procedure was validated against exhaustive enumeration on a reduced sample where all 2^n sign assignments are enumerable, giving an exact p of 0.00391 against a sampled 0.00420. Confidence intervals come from 20,000 paired bootstrap resamples of the same difference vector. Because nine comparisons are made against the reference within each metric and channel, all p-values are corrected by the Holm–Bonferroni step-down procedure, which controls the family-wise error rate without assuming independence between comparisons. Every statistic uses a fixed seed and is reproducible exactly.

Permutation was chosen over a paired t-test because it assumes only exchangeability under the null, not normality, and several of these difference distributions are visibly skewed — the GCN's first-death figures are bimodal across topologies. It was chosen over Wilcoxon signed-rank because it operates on the actual differences rather than their ranks, so the effect size and the test statistic are the same quantity.

F. Verification

Six gates were required to pass before any number in this paper was read. Energy conservation is asserted at the end of every single run to 10^{-9} J against the identity in §II-A.

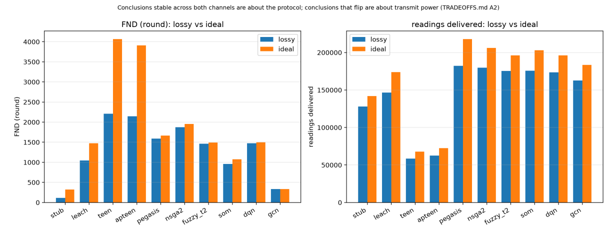

Fig. 5. 1

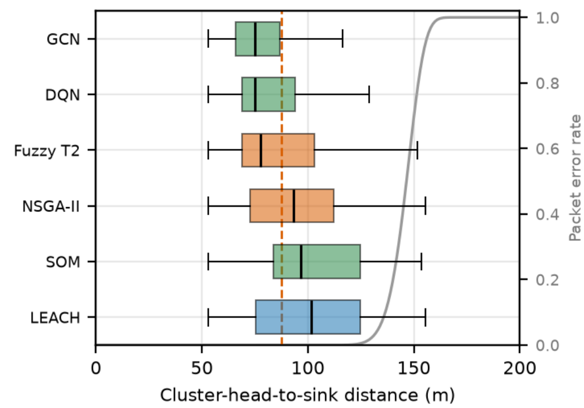

Fig. 6. 1

Radio energies are checked against hand-computed values on both sides of the crossover distance. A suite of invariants confirms that energy never goes negative, that dead nodes never transmit, that the living count is monotone, that the three death points are ordered, and that delivery ratios remain in range. Determinism is verified by running every protocol three times per seed and diffing all per-round and summary fields, excluding only the timing and memory fields that are not expected to be reproducible; all nine protocols reproduce exactly. A static audit of the protocol source confirms that no implementation writes to the energy or liveness arrays, calls the charging function, or chooses its own packet sizes — eighteen checks, all passing. Finally the aggregation pipeline is verified by recomputing one protocol's summary row independently and diffing at 10^{-9}.

A seventh gate exists specifically because of the lossy channel: the packet-error curve is emitted as a published artifact over 0–200 m and asserted to be non-decreasing, effectively zero at intra-cluster range, and strictly between zero and one somewhere in the band of sink distances. Had that last condition failed, the channel would be behaving as a binary range switch and the lossy results would carry no information.

The full sweep of 600 runs completes in 80 minutes on sixteen cores. All configuration constants live in a single frozen structure, no constant is defined anywhere else, and the raw per-round output of every run is retained rather than only the summaries.

TABLE IV

COST, LOSSY CHANNEL. COUNTED OPERATIONS ARE THE HONEST CROSS-PROTOCOL COMPARISON; WALL-CLOCK SETUP TIME REFLECTS IMPLEMENTATION QUALITY AS MUCH AS ALGORITHM DESIGN AND IS REPORTED SEPARATELY FOR THAT REASON. DQN AND GCN SETUP TIMES ARE INFERENCE ONLY: THEIR ONE-OFF OFFLINE TRAINING IS NOT INCLUDED HERE (SECTION **??**).

Protocol	Gen.	Latency (ms)	p95 (ms)	Control pkts ($\times 10^3$)	Ops per round	Setup (ms)	Peak mem (MB)
Direct (no clustering)	–	600	1,488	0.0	0	0.00	6.0
LEACH	G1	274	837	171.6	329	0.82	1.7
PEGASIS	G1	1,045	1,626	213.9	59	0.05	1.5
TEEN	G1	419	990	573.9	307	0.83	5.1
APTEEN	G1	407	979	554.3	296	0.80	5.1
NSGA-II	G2	387	573	10.8	86,348	102.89	2.1
Fuzzy T2	G2	401	1,028	9.7	2,301	5.14	1.6
SOM	G3	388	538	10.7	6,194	38.28	2.0
DQN	G3	402	918	11.9	8,210	0.52	1.7
GCN	G3	392	981	12.2	5,454	0.88	2.3

TABLE V

FIRST-NODE-DEATH DIFFERENCE AGAINST LEACH, IN ROUNDS, UNDER BOTH CHANNEL CONFIGURATIONS. PAIRED SIGN-FLIP PERMUTATION TEST (20 000 PERMUTATIONS) WITH PAIRED-BOOTSTRAP CONFIDENCE INTERVALS, HOLM-CORRECTED ACROSS THE NINE COMPARISONS WITHIN EACH CHANNEL. **FUZZY T2 AND DQN ARE SIGNIFICANT UNDER LOSS AND NOT SIGNIFICANT WITHOUT IT** — THE STUDY'S CENTRAL NEGATIVE RESULT.

	Lossy channel			Ideal channel		
Protocol	ΔFND	95% CI	p_{Holm}	ΔFND	95% CI	p_{Holm}
Direct (no clustering)	-932	[-944, -919]	0.0004	-1,153	[-1,162, -1,144]	0.0004
PEGASIS	+542	[+418, +647]	0.0004	+188	[+78, +293]	0.0090
TEEN	+1,163	[+1,109, +1,217]	0.0004	+2,593	[+2,542, +2,647]	0.0004
APTEEN	+1,099	[+1,055, +1,143]	0.0004	+2,434	[+2,392, +2,480]	0.0004
NSGA-II	+827	[+811, +844]	0.0004	+482	[+471, +493]	0.0004
Fuzzy T2	+417	[+393, +440]	0.0004	+17	[+0, +34]	0.1195
SOM	-87	[-135, -41]	0.0015	-399	[-443, -357]	0.0004
DQN	+426	[+404, +450]	0.0004	+22	[-4, +49]	0.1244
GCN	-710	[-764, -660]	0.0004	-1,140	[-1,193, -1,090]	0.0004

TABLE VI

LIFETIME, LOSSY CHANNEL, 30 PAIRED RUNS (MEAN ± SAMPLE STD, ROUNDS). FND/HND/LND ARE FIRST, HALF AND LAST NODE DEATH. AUC IS THE AREA UNDER THE LIVING-NODE CURVE, THE CENSORING-ROBUST SURVIVAL MEASURE.

Protocol	Gen.	FND	HND	LND	
Direct (no clustering)	–	114 ± 11	1,144 ± 183	3,202 ± 78	[illegible]
LEACH	G1	1,046 ± 43	1,679 ± 25	2,191 ± 81	167.1
PEGASIS	G1	1,588 ± 331	2,199 ± 9	2,630 ± 358	216.4
TEEN	G1	2,208 ± 164	5,601 ± 189	>6,999[c]	535.4
APTEEN	G1	2,144 ± 139	5,367 ± 161	>6,999[c]	518.4
NSGA-II	G2	1,873 ± 31	1,965 ± 23	2,697 ± 256	200.6
Fuzzy T2	G2	1,462 ± 50	2,006 ± 28	2,109 ± 48	191.9
SOM	G3	959 ± 127	1,998 ± 32	2,874 ± 115	196.8
DQN	G3	1,472 ± 55	1,920 ± 31	2,325 ± 59	192.1
GCN	G3	335 ± 142	1,824 ± 70	3,185 ± 142	180.2

[c] Censored: the network had not died at the 7000-round cap in any run. Use AUC, not LND, for these two.

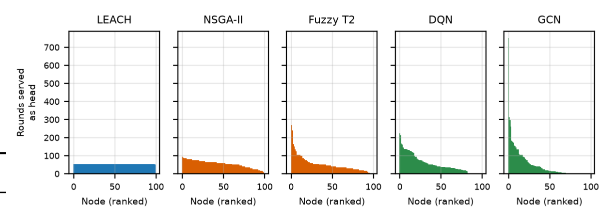

Fig. 7. 1

V. RESULTS

A. What Clustering Actually Buys

The direct-transmission baseline makes the first result concrete. LEACH reaches first node death at 1046 rounds against the baseline's 114, an improvement of 9.2×. But the baseline's last node dies at 3202 rounds against LEACH's 2191, 46% later. Clustering does not extend lifetime; it redistributes it, buying a much later first death at the cost of an earlier last one.

The mechanism is geometric. With the sink at 150 m and a crossover at 87.7 m, 36 of the 100 nodes lie close enough to transmit on the free-space branch and are individually cheap to operate. Under direct transmission those nodes are never asked to relay anyone else's traffic and simply outlive everything LEACH produces. Under rotation they subsidize the expensive far-field nodes.

This is why every lifetime claim in this paper is reported at three death points plus the area under the living-node curve. Any single figure silently picks a winner, and the two most commonly reported figures pick opposite ones.

The baseline also produces the clearest illustration of survivorship bias in delivery statistics. Its delivery ratio of 96.9%

TABLE VII
EVERY RANKING THAT MOVES WHEN THE CHANNEL CHANGES: 4 OF 40 PROTOCOL–METRIC RANKS. ALL ARE SINGLE-PLACE SWAPS BETWEEN ADJACENT PROTOCOLS. ORDERING IS THEREFORE STABLE; IT IS THE *significance* OF SPECIFIC FIRST-NODE-DEATH GAPS THAT IS NOT (TABLE V).

Metric	Protocol	Lossy rank	Ideal rank	Shift
hnd	Fuzzy T2	4	5	+1
hnd	SOM	5	4	-1
readings_delivered_total	Fuzzy T2	4	5	+1
readings_delivered_total	DQN	5	4	-1

is the highest of any configuration on the lossy channel, well above LEACH's 87.7%, and it is an artifact: the distant nodes that lose the most packets die first, so the surviving population is progressively closer to the sink. Measured directly, the baseline's delivery ratio rises from 81.6% over its first hundred rounds to 99.0% over its last hundred. Delivery ratio is never reported here without the living-node curve beside it.

B. Lifetime Across the Three Generations

Table III gives the full lossy-channel results. Against LEACH, with paired permutation testing and Holm correction across the family, every protocol differs significantly. The reactive pair leads on first death by a wide margin — TEEN at 2208 rounds, $\Delta = +1163$ — followed by NSGA-II at 1873 ($\Delta = +827$), PEGASIS at 1588, the DQN at 1472 and the fuzzy system at 1462. Two protocols are significantly worse than LEACH: the self-organizing map at 959 and the GCN at 335.

Three observations survive scrutiny and are worth stating separately from the ranking.

NSGA-II posts both the best non-reactive first death and the tightest spread of any protocol (standard deviation 31 rounds against LEACH's 43 and PEGASIS's 331). Its advantage holds in both channel configurations. But its convergence is modest, and we report that as a finding rather than suppress it: over a full run the population's best round-energy objective improves by 0.70% and its minimum-head-energy objective by 3.1%, while only cluster-size balance moves substantially, from 5.9 to 1.79. With five heads drawn from a hundred candidates, random initialization already lands near a good solution. NSGA-II's advantage here should therefore not be attributed to the sophistication of the search.

PEGASIS has by far the largest first-death variance of any protocol, 331 rounds against a typical 30–60. A greedy chain is sensitive to where the topology happens to place its outliers, and a single badly-positioned node changes the chain's whole cost structure. Its half-death figure, by contrast, has a standard deviation of 9 rounds — the tightest in the study — because once the chain has been rebuilt a few times its structure becomes topology-insensitive.

The GCN's first death is not merely poor but bimodal. Across the first five seeds it records 448, 394, 418, 112 and 98 rounds: on some topologies it behaves reasonably and on others a node drains almost immediately. A mean of 335 with a standard deviation of 142 describes neither mode. §V-D explains why.

TABLE VIII
SURVIVAL AND DELIVERY DIFFERENCES AGAINST LEACH UNDER BOTH CHANNELS. UNLIKE FIRST NODE DEATH (TABLE V), EVERY ONE OF THESE GAPS HOLDS IN BOTH CONFIGURATIONS, SO THESE ARE THE CONCLUSIONS THE STUDY CAN STATE WITHOUT A CHANNEL CAVEAT.

	Lossy		Ideal	
Protocol	Δ ($\times 10^3$)	p_{Holm}	Δ ($\times 10^3$)	p_{Holm}
AUC				
Direct (no clustering)	-35.1	0.0004	-32.8	0.0004
PEGASIS	+49.2	0.0004	+44.5	0.0004
TEEN	+368.3	0.0004	+389.6	0.0004
APTEEN	+351.3	0.0004	+371.7	0.0004
NSGA-II	+33.5	0.0004	+31.7	0.0004
Fuzzy T2	+24.8	0.0004	+22.1	0.0004
SOM	+29.7	0.0004	+29.2	0.0004
DQN	+25.0	0.0004	+22.5	0.0004
GCN	+13.1	0.0004	+11.1	0.0004
Readings				
Direct (no clustering)	-18.5	0.0004	-31.8	0.0004
PEGASIS	+35.6	0.0004	+44.2	0.0004
TEEN	-88.1	0.0004	-106.0	0.0004
APTEEN	-84.0	0.0004	-101.3	0.0004
NSGA-II	+33.2	0.0004	+32.2	0.0004
Fuzzy T2	+28.8	0.0004	+22.4	0.0004
SOM	+29.2	0.0004	+29.3	0.0004
DQN	+26.9	0.0004	+22.5	0.0004
GCN	+16.2	0.0004	+9.8	0.0004

C. A Conclusion the Channel Sweep Retracts

Applying the decision rule of §IV-B to the 30-run sweep, one conclusion does not survive, and it is a load-bearing one. On the lossy channel the fuzzy system and the DQN improve first death over LEACH by 417 and 426 rounds respectively, both far below the corrected significance threshold. On the ideal channel the same comparisons give 17 and 22 rounds at $p = 0.12$ and $p = 0.12$ — not significant. Remove packet loss and both protocols become statistically indistinguishable from LEACH on the metric this literature most often reports.

We measured the mechanism rather than arguing it. Splitting each protocol's energy by accounting category over a single seed and reading off the mean head-to-sink distance in the same run:

	retry energy	control energy	mean head→sink	total
LEACH	6.39%	12.73%	101.8 m	68.02 J
SOM	5.28%	2.38%	102.8 m	55.79 J
Fuzzy T2	3.65%	2.33%	85.6 m	58.09 J
NSGA-II	3.57%	2.46%	94.2 m	54.46 J
DQN	2.31%	3.95%	81.1 m	56.44 J
GCN	2.25%	3.85%	77.2 m	55.41 J

LEACH's average cluster head sits at 101.8 m — past the 87.7 m crossover, on the fourth-power branch, and far enough out to be losing packets — and it spends 6.4% of its entire energy budget on retransmissions. The DQN's average head sits at 81.1 m, inside the free-space branch, and spends 2.3%. That four-point gap is the whole of the first-death advantage.

Delete it by making the channel lossless and the advantage disappears.

The honest claim is therefore narrower, and more interesting, than "the soft-computing and learned protocols cluster better." Their demonstrable skill is keeping the head-to-sink hop inside the free-space regime, and that skill pays only in proportion to how lossy the hop is. The same table shows LEACH's compensating strength: it consumes 20% more total energy than the DQN yet matches its first death on a clean channel, because its epoch rule rotates head duty perfectly.

The retraction is specific to first death and we do not overstate it. On area under the living-node curve and on readings delivered, every protocol's difference from LEACH is significant at the corrected floor in both channels, in the same direction and at similar magnitude — the fuzzy system gains 24,793 node-rounds on the lossy channel and 22,120 on the ideal. The correct statement is that the advantage in overall survival and in data delivered is robust to the channel, and only the advantage in first death is not. First death is a single order statistic and the most fragile of the four lifetime measures.

Across all forty protocol-metric rankings, four ranks move between channel configurations, and both moving pairs are near-ties.

D. What Temporal Credit Assignment Buys

The DQN and the GCN consume identical inputs and differ only in architecture and training signal, so the gap between them isolates one variable. The DQN posts the tightest first-death distribution of any protocol in the study, 1404 to 1530 rounds across seeds, a spread of 126. The GCN's spans 350.

Measured over a 400-round run, the GCN uses 79 distinct cluster heads out of 100, with its busiest node serving 106 of the 2000 available head-rounds against a uniform share of 20 — roughly five times over-service. LEACH, by construction, uses all 100 nodes exactly 20 times each.

The cause is structural rather than a defect in training. The GCN is trained against a single-round energy-balance objective with no temporal credit assignment; it cannot represent "sacrifice this round to extend lifetime." Its fourth input feature lets it observe rotation state, but nothing in its loss rewards acting on it. The DQN, with the same four features and a discount factor of 0.999, has exactly that machinery. This is the study's cleanest single measurement of what temporal reasoning contributes to clustering, and it supports the framing the two protocols were chosen to test: the graph network performs spatial and structural reasoning, the Q-network performs temporal reasoning, and for a metric defined by the first failure, the temporal half is what matters.

The self-organizing map is the instructive negative case. It shares LEACH's head-placement problem — a mean head-to-sink distance of 102.8 m, the worst in the study — without LEACH's rotation guarantee, and it is the only centralized protocol to lose to LEACH on first death in both channel configurations.

TABLE IX
WHERE THE ENERGY GOES AND WHERE THE HEADS SIT, MEASURED BY THE ENGINE'S OWN ACCOUNTING CATEGORIES (SEED 0, 1100 ROUNDS). PACKET ERROR IS NEAR ZERO BELOW 120 M, SO THE *mean* HEAD DISTANCE CANNOT EXPLAIN THE RETRY GAP — EVERY MEAN IS BELOW THAT. THE TAIL CAN: ACROSS THE SIX SINGLE-HOP PROTOCOLS, THE SHARE OF HEAD-ROUNDS BEYOND 120 M PREDICTS RETRY ENERGY SHARE WITH $r = 0.95$. PEGASIS IS THE EXCEPTION THAT PROVES THE RULE: ITS LEADER ROTATES UNIFORMLY AND 30% OF LEADER-ROUNDS SIT BEYOND 120 M, YET IT RETRIES BARELY AT ALL, BECAUSE ONLY THE LEADER EVER FACES THE SINK.

Protocol	Retry (%)	Control (%)	Head-to-sink (m) mean	Head-to-sink (m) p90	Heads >120 m (% of rounds)
LEACH	6.39	12.73	102.1	143.5	29.9
PEGASIS	1.39	4.38	102.2	143.5	30.0
SOM	5.28	2.38	102.8	136.4	31.1
NSGA-II	3.57	2.46	94.2	127.5	17.7
Fuzzy T2	3.65	2.33	85.6	122.1	11.5
DQN	2.31	3.95	81.1	111.2	5.1
GCN	2.25	3.85	77.1	104.6	3.2

E. Delivery and the Reactive Tradeoff

TEEN and APTEEN dominate every lifetime metric and fail to reach full depletion within the 7000-round cap in all 30 trials, so their last-death figures are lower bounds. Their area under the living-node curve is 3.2× LEACH's. This is not free. Their data yield is 0.109 and 0.121 against roughly 0.9 for the proactive protocols, and they deliver 58,551 and 62,579 readings against LEACH's 146,616. They survive by not reporting, and their reporting rate — 0.15 and 0.17 transmissions per living node-round against 1.0 for everything else — states the same fact from the other direction. Whether that is a good trade depends entirely on the application, which is precisely why both quantities are reported and neither is summarized into a single score.

PEGASIS delivers the most readings of any protocol, 182,179, or 24% more than LEACH, while receiving the fewest packets at the sink, 2272. Its aggregation ratio is 80 readings per delivered packet against LEACH's 13.3. On a packet-count basis it appears to be the worst performer in the study by a factor of five; on a readings basis it is the best. Both figures are in Table III, and only the second is a delivery comparison.

Delivery ratios span a narrow band on the lossy channel, 84.2% to 91.4%, and rise to 98.8–99.9% on the ideal channel, confirming that packet loss and not protocol logic accounts for the difference.

F. Latency and Computational Cost

Latency separates the protocols structurally. The clustered protocols cluster around 390–420 ms mean, LEACH is lowest at 274 ms because its stochastic election produces smaller average clusters, and PEGASIS is an outlier at 1045 ms mean and 1626 ms at the 95th percentile. Chain depth is the dominant latency cost and no amount of energy efficiency compensates for it. The ordering is structurally correct, but the absolute figures are lower bounds, since the model contains no queueing or contention delay.

TABLE X

CELLS ORDERED BY NODE DENSITY. IF DENSITY DROVE LIFETIME THE ROWS WOULD TREND WITH IT; THEY DO NOT — THE NUMBERS TRACK THE MEAN SINK DISTANCE COLUMN INSTEAD. AVERAGED OVER ALL TEN PROTOCOLS, TRIPLING THE NODE COUNT CHANGES FIRST NODE DEATH BY A FACTOR OF 0.88–1.26, WHILE TRIPLING THE FIELD SIDE CHANGES IT BY A FACTOR OF 5–7. THE TWO CELLS CLOSEST IN DENSITY (50 AND 44 NODES/HA) DIFFER BY 1.7× FOR NSGA-II, 2.1× FOR LEACH AND 5.5× FOR DIRECT TRANSMISSION. NOTE ALSO THAT *clustering only pays once the sink is far*: IN THE 50 × 50 M ROWS THE NO-CLUSTERING BASELINE OUTLIVES LEACH.

N	Field (m)	Density (nodes/ha)	Mean sink dist. (m)	First node death (rounds)		
				Direct	LEACH	NSGA-II
50	150×150	22	156	23	155	717
100	150×150	44	156	22	303	1,052
50	100×100	50	104	120	641	1,818
150	150×150	67	156	22	354	1,077
100	100×100	100	104	112	1,034	1,872
150	100×100	150	104	111	1,085	1,865
50	50×50	200	52	2,178	1,791	2,184
100	50×50	400	52	1,943	1,617	2,173
150	50×50	600	52	1,677	1,586	2,160

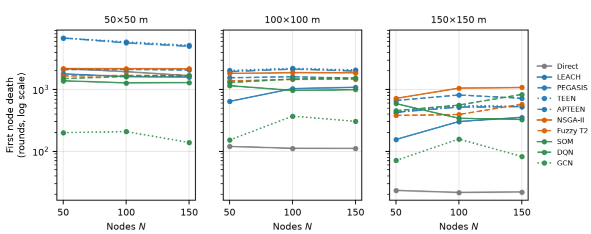

Fig. 8. 1

Computational cost is where the three measurement notions of §IV-C disagree most sharply, which is the argument for keeping them separate. NSGA-II costs 102.9 ms per round of setup against LEACH's 0.82 ms, a factor of 126, and 86,348 counted operations against 329, a factor of 262. The self-organizing map is second at 38.3 ms. But the ordering by counted operations and by wall-clock is not the same: the DQN performs 8210 counted operations per round in 0.52 ms, while the fuzzy system performs 2301 in 5.14 ms. The DQN does 3.6 times the algorithmic work in a tenth of the time, because its work is a dense matrix product and the fuzzy system's is an iterative type-reduction loop. Reporting either number alone would invert the comparison.

Control traffic differs by more than an order of magnitude across classes — 573,870 packets for TEEN and 171,644 for LEACH against 9657 to 12,216 for the centralized five. That gap is substantially an artifact of the accounting model rather than of clustering quality, and Section VI quantifies it.

THREATS TO VALIDITY

Every result in Section V depends on choices that a reader cannot see in the numbers. This section states them, ordered by how far each could move a conclusion, and it names the one that could move the most first.

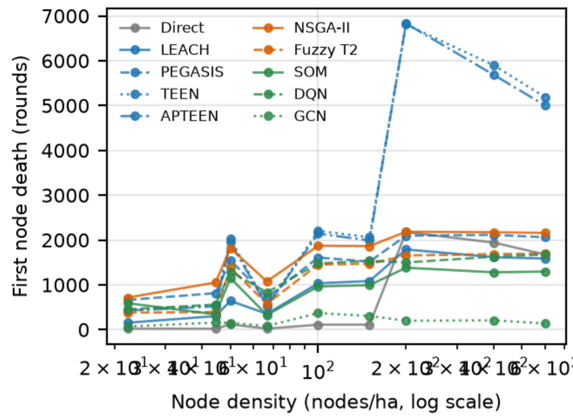

Fig. 9. 1

A. The Control-Traffic Subsidy

Centralized protocols are charged nothing for the uplink that informs the sink of node state. This is the largest single confound in the study and it favours five of the ten configurations.

The justification is quantitative. A separate 200-bit status report per node per round, over a sink distance above 100 m and therefore on the fourth-power branch, costs roughly 3.6×10^{-5} J per node per round, accumulating to about a quarter of the entire 1 J budget across the round cap. Charging it would sink all five centralized protocols for a reason unconnected to whether their clustering decisions are any good, and the piggyback assumption is standard in the centralized-clustering literature. But a defensible modelling choice is still a modelling choice, and its magnitude belongs in the results rather than in a footnote.

Measured over the full sweep, LEACH spends 12.73% of its consumed energy on control traffic against 2.33% for the fuzzy system, 2.46% for NSGA-II, 2.38% for the map, 3.95% for the DQN and 3.85% for the GCN. The subsidy is therefore worth roughly nine to ten percentage points of the total budget. In packet terms the gap is starker still — 171,644 control packets for LEACH and 573,870 for TEEN against 9657 to 12,216 for the centralized five — but the packet gap overstates the effect, since control packets are a fifth the size of data packets and most are short-range.

Two consequences follow. First, a claim of the form "NSGA-II extends first death by 80% over LEACH" conflates clustering quality with an accounting convention and should not be made without qualification. Second, the reliable comparisons in this paper are the within-class ones: centralized against centralized, distributed against distributed, with the chain protocol comparing cleanly to neither. Section V's strongest results — the channel-conditional retraction, the DQN-versus-GCN rotation contrast — are all within-class and are unaffected.

The clean resolution is an ablation that charges the uplink honestly and re-runs the sweep. It would convert this caveat into a measurement, and it is the single most valuable follow-

TABLE XI
First node death (rounds) across node counts and field areas, lossy channel, 15 paired runs per cell.

Protocol	Gen.	50×50 m N=50	50×50 m N=100	50×50 m N=150	100×100 m N=50	100×100 m N=100	100×100 m N=150	150×150 m N=50	150×150 m N=100	150×150 m N=150
Direct (no clustering)	–	2,178	1,943	1,677	120	112	111	23	22	22
LEACH	G1	1,791	1,617	1,586	641	1,034	1,085	155	303	354
PEGASIS	G1	2,097	2,114	2,062	1,550	1,610	1,517	663	815	720
TEEN	G1	6,808	5,899	5,186	2,028	2,201	2,064	425	550	532
APTEEN	G1	6,831	5,684	5,002	1,947	2,141	1,992	427	519	527
NSGA-II	G2	2,184	2,173	2,160	1,818	1,872	1,865	717	1,052	1,077
Fuzzy T2	G2	1,656	1,685	1,698	1,379	1,454	1,470	381	393	577
SOM	G3	1,382	1,280	1,296	1,153	965	994	591	342	328
DQN	G3	1,500	1,628	1,677	1,296	1,478	1,537	454	559	833
GCN	G3	200	208	138	152	371	305	71	158	82

The base station scales with the field at $(W/2, 1.5W)$, so field size sets channel severity as well as area. Node-to-sink distance by field size — 50×50 m: mean 52 m, 0% of nodes beyond 120 m; 100×100 m: mean 104 m, 33% of nodes beyond 120 m; 150×150 m: mean 156 m, 75% of nodes beyond 120 m. At 50 × 50 m no link reaches the error waterfall at all, which is why the advantage of the optimized and learned protocols over LEACH collapses there.

TABLE XII
Head-duty concentration over 1100 rounds (seed 0), in rounds served as cluster head per node. LEACH's 55 is exact rather than approximate: its epoch mechanism elects every node exactly once per 20 rounds. GCN's busiest node serves 13.6 times as often, which is the direct cause of its early first node death (Table VI) — it does not choose badly, it chooses the same good node until that node dies.

Protocol	Busiest node	Median	Never served	Gini
LEACH	55	55	0	0.001
PEGASIS	11	11	0	0.000
NSGA-II	95	60	0	0.207
Fuzzy T2	360	40	4	0.416
SOM	175	50	10	0.393
DQN	223	38	16	0.476
GCN	750	16	13	0.700

up this codebase supports. It has not been run.

B. Transmit Power Is Bracketed, Not Swept

Section V-C treats the channel comparison as a result, which it is, but the design bounds what it establishes. Two configurations bracket a continuum; they do not trace it. The lossy setting places the sink hop near the packet-error waterfall and the ideal setting removes loss entirely, so a conclusion stable across both is stable across that interval — but a transmit power between them, or above the lossy setting, would produce intermediate behaviour this study does not measure. A three- or four-point sweep over transmit power would trace the curve properly and would show where the fuzzy system's and the DQN's first-death advantage over LEACH actually crosses the significance threshold. That sweep has not been run.

Related parameters are equally unspecified in the source material and equally consequential: the path-loss exponent, the shadowing standard deviation, the noise floor, and the retry budget. We used conventional values for 2.4 GHz outdoor short-range links and published the resulting packet-error curve so that the choice is auditable, but we did not vary them.

C. The Pre-Training Asymmetry

The DQN and the GCN arrive pre-trained; the other seven protocols do not. Three distinct caveats follow and all belong in any citation of these results.

Their reported setup time and operation counts are inference costs only. Training is real, is paid once, and is reported separately. It must not be read as free.

The seven non-learned protocols receive no equivalent offline preparation. This is a fair comparison of a deployed system and it is not an apples-to-apples algorithmic comparison. Both statements are true simultaneously, and which one matters depends on the question being asked.

Generalization is demonstrated only across topologies drawn from the same distribution used in training — uniform random placement, one hundred nodes, the same field and sink position. The evaluation seeds are genuinely disjoint from the training seeds, so the result is not memorization, but nothing here supports a claim that either policy transfers to a different density, field size, or sink placement.

D. Modelling Simplifications

The radio model accounts for transmit, receive and fusion energy and nothing else. There is no idle-listening cost, no sleep-wake transition energy, no radio startup, and no processor energy. Idle listening frequently dominates real deployments, and including it would compress the differences between all ten configurations toward zero. Energy is a linear reservoir with no rate-capacity effect, no recovery effect, and no voltage cutoff.

Fusion is ideal. A head that receives any number of member packets emits exactly one, so compression is lossless and free in bits. This systematically favours high-fan-in structures, and PEGASIS most of all: its aggregation ratio of 80 readings per delivered packet is exactly the quantity this assumption idealizes. The engine enforces the rule symmetrically, so no protocol can exploit it beyond what its own topology allows, but the assumption still shapes the chain-versus-cluster comparison more than any other single simplification.

There is no MAC layer. Scheduling is assumed perfect and collision-free, with no contention, no hidden terminals, no inter-cluster interference and no capture effect. Packet loss comes only from distance-dependent error. Adding interference would penalize dense-cluster protocols relative to the chain and would make the number of simultaneously active heads a variable this study cannot see. Latency inherits the same limitation: it is analytic rather than queueing-based, retransmissions cost energy but not modelled time, and the absolute figures in Section V-F are lower bounds whose ordering is meaningful and whose magnitudes are not.

Everything is static. Nodes never move and fail only by depletion, so per-link shadowing is drawn once and held fixed. There is no mobility, no node addition, and no environmental change across the run. The head-to-sink hop is single-hop for nine of ten configurations, matching each protocol's original specification, which leaves the chain as the sole multi-hop contrast.

Sensor data is a mean-reverting process rather than a real trace. The reactive protocols' results are conditional on that model, and a bursty or spatially correlated field would change them substantially. The thresholds were set against the process's stationary distribution before any result was observed, and were not adjusted afterward.

Finally, every protocol targets the same nominal head count but realizes it differently, and LEACH's realized count is deliberately left stochastic. Part of every observed difference is therefore attributable to realized cluster count rather than to selection logic. Fixing the target isolates the question these protocols actually differ on — given roughly five heads may be chosen, which five — but it does not eliminate the effect.

E. Scope

This study establishes results at one network scale, one energy budget, and one traffic model. There is no scalability sweep, and three protocols have complexity superlinear in node count. Node energy is homogeneous; the configuration supports heterogeneity but it is disabled throughout. Every node senses every round, with no duty cycling, event-driven traffic, or query workload — APTEEN's query facilities in particular are out of scope, and only its reporting-frequency mechanism is modelled. There is no hardware testbed and no measured trace; every number is simulation output under the assumptions above.

F. On Not Calibrating

No protocol was tuned to reproduce a published figure. Only protocol descriptions, the radio model, and simulation parameters were taken from prior work; its numeric results were discarded before implementation began. Disagreement with published tables is therefore expected and is not evidence of error here.

The one place this commitment was tested is documented in Section III-D. The DQN's discount factor and training length were revised after an observed divergence, after which a stopping rule was fixed in advance and the result recorded as final regardless of outcome. We state this because the alternative — presenting the final hyperparameters as if they had been the first — is the ordinary way such revisions disappear from a methods section.

CONCLUSION

We built a single simulation engine in which nine clustering protocols spanning twenty-five years of design practice execute under provably identical physical conditions, and evaluated them over 600 runs across two channel configurations. The engine owns energy accounting, packet sizing, fusion charges, the channel, and reading accounting; a protocol returns only a cluster structure. Fairness is therefore a property of the construction rather than of the authors' care, and it is verified by static audit rather than asserted.

Four findings hold up.

Clustering does not extend network lifetime so much as redistribute it. LEACH delays first node death by a factor of 9.2 over direct transmission while bringing last death forward by nearly a third, because a third of the nodes lie close enough to the sink to transmit on the free-space branch and are penalized rather than helped by rotation. No single lifetime figure summarizes this, and the two most commonly reported figures order the protocols in opposite directions.

The advantage of the soft-computing and learned protocols over LEACH in first node death is conditional on the channel. Under packet loss the fuzzy system and the Q-network improve on LEACH by 417 and 426 rounds; with loss removed the same comparisons give 17 and 22 rounds and lose significance. Direct measurement of the energy split shows why: LEACH's average head sits at 101.8 m, past the radio crossover and inside the packet-error waterfall, spending 6.4% of its budget on retransmissions against the Q-network's 2.3% at 81.1 m. The demonstrable skill of the newer protocols is keeping the sink hop inside the free-space regime, and it pays in proportion to how lossy that hop is. Their advantage in overall survival and in readings delivered is not conditional; it holds in both configurations at the corrected significance floor.

Temporal credit assignment is worth measuring separately from representational capacity. The Q-network and the graph network consume identical features and differ only in architecture and training signal. The Q-network produces the tightest first-death distribution in the study; the graph network produces a bimodal one, over-serving its busiest node fivefold, because a single-round objective cannot represent deferring cost to extend lifetime even when rotation state is visible in its inputs.

Reactive reporting buys lifetime at a price that must be reported alongside it. TEEN and APTEEN survive past the round cap in every trial with an area under the living-node curve more than three times LEACH's, at a data yield of 0.11 to 0.12 against roughly 0.9 for the proactive protocols.

The methodological point generalizes beyond these nine protocols. Running the entire study at both endpoints of an unspecified parameter, and committing in advance to report only what survives both, retracted a comparison that either endpoint alone would have supported. Simulation studies routinely fix such parameters silently; where the parameter sits near a nonlinearity, as transmit power does relative to the packet-error waterfall, the choice can decide the ranking. The cost of bracketing it was a doubled runtime, which at this scale is eighty minutes.

The largest confound remaining is the accounting subsidy granted to centralized protocols, worth nine to ten percentage points of the energy budget. An ablation that charges the uplink honestly would convert it from a caveat into a measurement, and is the most valuable extension of this work. Sweeping transmit power across several intermediate values, and repeating the study across network scales, would establish how far these results extend beyond the single operating point measured here.

All configuration constants, per-round outputs, verification artifacts, and the packet-error calibration curve are released with the code.

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